

When an 8-K meets unusual option skew

Gator Quant Hacks 2026 · Systematic Trading · Massive

1. Hypothesis

An 8-K tells investors that a company has publicly disclosed something. We test whether unusual option prices before a filing provide a useful clue, and whether the filing separates those moments from ordinary days. The filing supplies context, not a directional forecast.

Portfolio-manager hypothesis. A pre-filing call-versus-put IV residual may capture relative repricing around information about to become public. Conditioning on an actual disclosure could distinguish those episodes from routine option noise. A same-company placebo tests whether the filing adds information beyond skew alone; the null is a zero net event-minus-placebo return after costs.

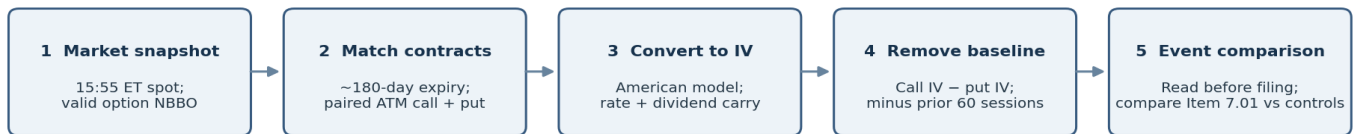
Item 7.01 is the Regulation FD section of Form 8-K. Companies may use it to make information broadly public after selective communication; its varied content is not inherently bullish or bearish. [Investor.gov](https://www.investor.gov) explains the category.

2. Method: IV180 construction

At 15:55 ET, use spot from the last completed five-minute bar. Select a standard expiry nearest 180 calendar days within 150–210 days and the nearest strike with both call and put. Use valid bid/ask midpoints no more than 60 seconds old. QuantLib solves American IV with prior Treasury yields and trailing paid dividends as carry.

IV is the model volatility matching an option price, not observed trader positions. Subtract each company's prior 60-session average call-minus-put IV (minimum 30 valid observations), then standardize across companies. High skew uses $z > 0.43$, a nominal normal-reference cutoff. Historical OI is unavailable; volume is not substituted for it.

From option quotes to a pre-filing signal



IV180 is a maturity target, not an exact 180-day contract. The IV quotes form the signal; daily option bars mark trade outcomes.

Figure 1. IV quotes form the predictor. Daily option bars later mark modeled trade outcomes.

3. Results

The reproduced filing/signal window is Mar 7, 2022–Dec 31, 2024; exit marks are capped at the same end date. Protective-put means are 0.73% and 1.25% of entry spot at 10 and 21 sessions. Matched high-skew placebo means are 0.34% and 0.69%. The gaps are 0.39 and 0.56 percentage points, with t-statistics 0.67 and 0.61: positive but statistically weak.

Monthly cohort measure	Reproduced value
Annualized Sharpe (33 cohorts)	0.42
Summed P&L / cohort-index drawdown	10.85% / 6.54%
Positive cohorts	42%

Risk reversal averages 0.93% at 21 sessions, above four standard structures but below protective put at 1.25%. The protective-put event gap remains negative in 2024. Cohort Sharpe is separate from trade P&L and is not a complete self-financing account return.

Reproduced 2022–2024 in-sample results · 2% option-premium haircut

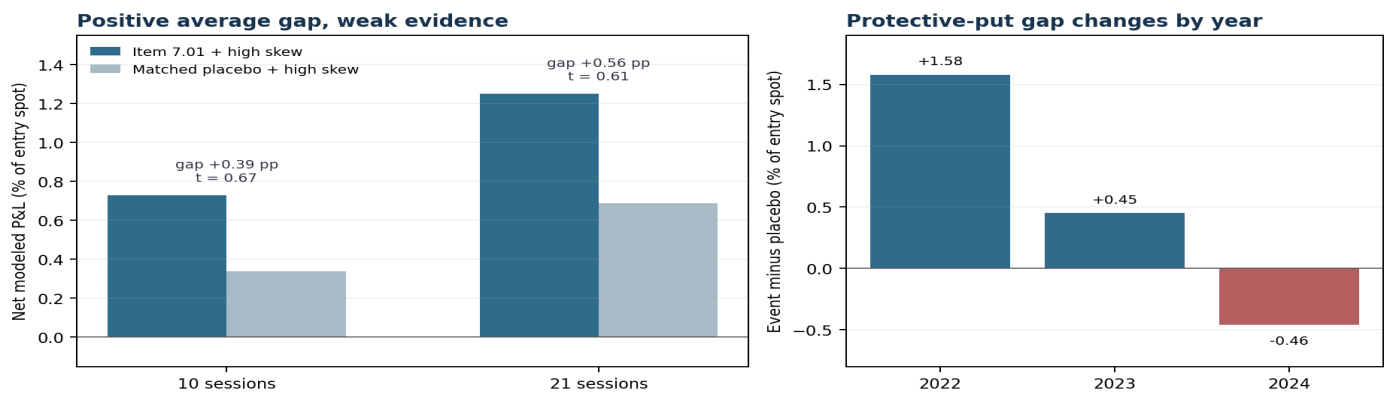


Figure 2. Event/control means and year gaps. The positive pooled effect is uncertain and varies by year.

4. Risks and failure conditions

The sample is small, the static universe can have survivorship bias, and daily option bars are imperfect fill proxies. Put premium can erode returns in quiet markets. Costs may miss spreads and market impact. Sharpe 0.42 is below the requested target; persistent profitability is not established.

5. Trade and reproduction

For high-skew Item 7.01 filings, enter on the first session after filing: 100 shares plus one 3–6 month put about 5% below spot; exit after 10 sessions. Charge 2% of option premium per leg per side and 3 basis points per stock side. The notebook tests five standard and four advanced structures. Full local reproduction used the original IV input and bounded option data; a fresh API-builder day matched both IVs to six decimals. Evaluation stays disabled. Expiry, entry-timing and category sensitivity remain pending.